Performance Enhancement in Full-Duplex AF Relay System through Smart Antenna Allocation

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***Abstract*—Energy efficiency (EE) and spectral efficiency (SE) are two crucial performance metrics for next-generation wireless standards. In this work, the SE and EE of a full-duplex (FD) relay system based on the amplify-and-forward (AF) scheme is investigated. Specifically, it has been shown that residual self-interference (RSI) due to imperfect self-interference cancellation (SIC) impacts the SE and EE of the FD relay system. A smart antenna allocation (AA) scheme has been proposed that enhances the SE and the EE of the FD relay system through optimally allocating the antennas at the FD relay. Further, the impact of relay location has also been studied and it has been shown that the SE and EE of the FD relay systems can be maximized through optimal AA scheme and relay location. The proposed analysis is corroborated through simulations, which are in good conformity with the analytical results.**

***Index Terms*— Full-duplex, spectral efficiency, energy efficiency, residual self-interference, amplify-and-forward, antenna allocation.**

I. Introduction

The ever-growing quest for ultra-reliable high data rate services has boosted the consumption of wireless data manifold [1]. The advent of high definition (HD) on-demand videos, real-time online games and other multimedia applications would aggravate the global wireless traffic additionally. Recently, it has been found that the average monthly usage per smartphone in India is around 10 gigabytes (GB), which is highest in the world [2]. Further, the total mobile data traffic will continue to rise globally, and as per recent estimation, it is predicted to reach 131 exabytes (EB) per month at the end of 2024 [2]. In order to sustain the anticipated growth, the next-generation of wireless standard, viz. the fifth-generation (5G), is envisioned to provide 1000 times more throughput than the existing fourth-generation (4G) systems [3]. Full-duplex (FD) communication has emerged as a promising technique for increasing the spectral efficiency (SE) of the next-generation wireless system. Theoretically, FD has the potential to double the SE of the wireless system by utilizing the same time-frequency resource for concurrent transmission and reception [4].

The demand for higher throughput and large SE results in massive energy consumption in the wireless networks. As the capacity of a battery in wireless terminals is limited, there is a persistent demand for higher energy-efficient systems. Moreover, the increasing energy consumption in wireless networks inevitably leads to a considerable carbon footprint, which significantly contributes to environmental pollution [5]. As estimated in a recent study [6], around 1500 tera-watt hour power is consumed by the global information and communication technology (ICT) ecosystem, which accounts for 5% of the total CO_2 emissions globally. Hence to support the predictable traffic sustainably, the ambitious goal for the next-generation wireless standard is to scale down the energy consumption upto 100 times [1]. Various techniques proposed in the literature to support above and for energy-efficient wireless systems are surveyed in [7]–[9].

Numerous studies have shown that relaying techniques enhance the coverage area and improve the performance of wireless systems. Incorporating FD communication with relaying techniques has been shown to further improve the performance of wireless networks. For instance, by exploiting the benefits of FD and relaying together, the efficiency of the cellular networks can be improved significantly [10]. This merger of FD and relaying systems is commonly referred to as the FD relay system. In the FD relay system, the FD relay transmits and receive over the same time-frequency resource rather than two orthogonal time-frequency slot as in half-duplex (HD) relay system. In an amplify-and-forward (AF) relay system, the received signal from the source is amplified and forwarded by the relay to the destination node. Since the AF relaying only amplification is performed, it is much less complex than the decode-and-forward (DF), which is generally a regenerative technique. In [11], a FD AF relay system is optimized for minimizing symbol error rate (SER) in terms of power allocation and relay location. Authors have optimized power allocation and relay position for the FD DF relay system for minimizing the outage probability in [12]. FD relaying has also been explored for mitigating the impact of higher path-loss in the millimeter waves communication [13].

The self-interference (SI) due to leakage from the transmit antennas to the receive antennas at the FD relay node limits the performance of FD relays. To avail the true benefits of FD relays, a significant challenge is to mitigate the impact of strong SI. Various techniques for suppressing SI has been reported in the literature and comprehensively surveyed in [14]. Usually, SI is suppressed through passive techniques such as antenna separation and isolation, which is followed by active analog/RF and digital cancellation. Due to the various transmit imperfections, complete SI cancellation is not

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practically feasible. This remaining residual SI RSI usually impacts the performance of the FD relay system. The impact of RSI on the EE performance for the FD AF relay system is analyzed in [15]. Precoding design for AF based FD multiple-input multiple-output (MIMO) relay affected by RSI is studied in [16]. Further in [17], authors have analyzed SE performance for partial FD AF relaying with RSI. The error performance for FD AF relaying is also studied in [10], where the authors have investigated the diversity order under RSI.

Motivated by the above, the proposed work investigates the SE and EE of FD based AF relay system. As shown in the literature, the RSI impacts the information reception, i.e., the source-to-relay link; however, the information transmission in the FD relay system is interference-free, i.e., the relay-to-destination link is not affected by the RSI. The above scenario can be exploited through a smart antenna allocation (AA) scheme wherein the antennas are optimally allocated based on maximizing the performance of the FD relay system. As compared to the conventional equal antenna distribution for information reception and transmission at the FD relay, allocating more number of antennas for the information reception improves the SE of the FD relay system. Thus, the EE of the FD system also improves. Further, due to the reduced number of RF transmit chains for information transmission, the circuit power consumption is also reduced, which provides additional gain in the EE of the FD relay system.

II. System Model

As shown in Fig. 1, we consider a two-hop relay system with a source (S) communicating to the destination (D) via AF relay (R). The source-to-relay distance and the relay-to-destination distance are denoted as D_{SR} and D_{RD}, respectively, whereas D_{SD} denotes the source-to-destination distance. The sum distance of source-to-destination via relay link is D, i.e., $D = D_{SR}+D_{RD}$. We consider a scenario where the direct link between source and destination is obstructed or shadowed heavily, and the relay is deployed for coverage extension. Since there is no direct link, $D > D_{SD}$, i.e., relay cannot be placed at the source-to-destination line. The relay node is equipped with multiple antennas, whereas the source and destination are presumed to be a single-antenna devices[1]. The number of antennas at the relay node is N_R. The relay is operating in the full-duplex mode by transmitting and receiving in the same time-frequency resource, whereas the source and destination are operating in half-duplex mode.

A. Channel Model

The channel between source-to-relay $\mathbf{h}_{SR}$ and relay-to-destination $\mathbf{h}_{RD}$ both are modelled as frequency non-selective Rayleigh flat fading channel with channel elements $h_i \sim \mathcal{CN}(0,1)$ being independent and identically distributed (i.i.d.).

[1]This is specifically true for cellular networks wherein multi-antenna base station (BS) acts as a relay for single antenna mobile users. Further, it can be noted that the BSs have ample space to accommodate multiple antennas, whereas the user equipment is severely constrained by space and complexity requirements.

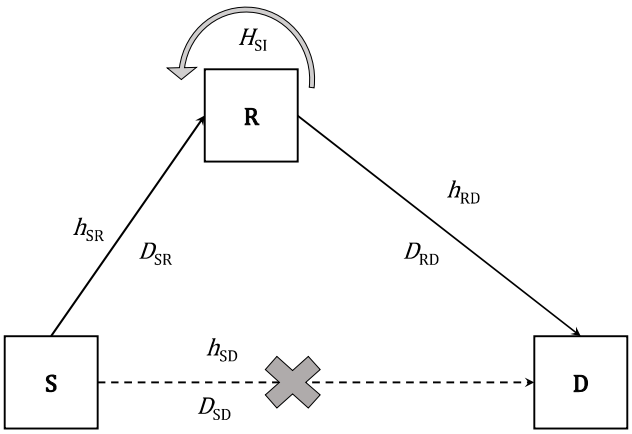

Fig. 1. Schematic for the FD Relay System.

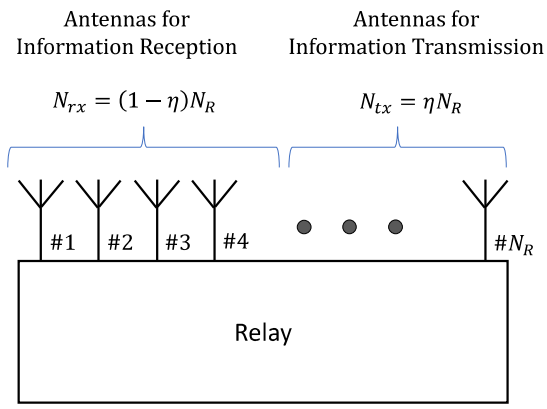

Fig. 2. Schematic for Antenna Allocation at the FD Relay.

Since the direct link between the source and destination is assumed to be heavily obstructed, i.e., $h_{SD} = 0$, hence the transmission occurs only via relay. Furthermore, channel state information (CSI) is assumed to be perfectly known at the FD relay [16]. s_x and $\mathbf{r}_x$ denotes the transmit data symbol from the source and the relay respectively. The elements of transmitted data symbols are assumed to be i.i.d. with zero mean and unit variance. The self-interference channel $\mathbf{H}_{SI}$ is also modeled as frequency non-selective Rayleigh flat fading channel denoted by i.i.d. elements, $h_{i,j} \sim \mathcal{CN}(0,1)$ [18].

B. Antenna Allocation Scheme

As illustrated in Fig. 2, the N_R antennas at the relay node can be distributed for information transmission and reception. The distribution of N_R antennas at relay can be done on the basis of parameter η which is termed as antenna allocation factor. So $N_{tx} = \eta N_R$, antennas at the FD relay transmit the information to the destination while the remaining $N_{rx} = N_R - N_{tx} = (1-\eta)N_R$ antennas at the FD relay are used for information retrieval from the source. Based on η, an optimal

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antenna allocation scheme is proposed which maximize the SE and EE of the FD relay system. The performance of this optimal AA scheme is compared with the traditional equal AA scheme, where the antennas at the relay are allocated in equal number for information transmission and reception i.e., $\eta = 0.5$.

C. Residual SI Model

The residual SI (RSI) has been characterized in two ways in the literature, viz. fading model and complex Gaussian random model. In the fading model, RSI is characterized by a statistical fading model like Rayleigh/Rician fading [19]. In contrast, the complex Gaussian random model characterizes the RSI as a Gaussian random variable with its variance depending on the underlying SIC methods. The Gaussian model has also been confirmed through practical measurements and has been widely used in literature due to its generality in modeling the RSI [10], [20]–[22]. In the present work, RSI is characterized by a complex Gaussian model owing to its generality in studying the impact of RSI.

The received signal $\mathbf{r}_y$ at the relay can be written as:

$$\mathbf{r}_y = \mathbf{h}_{SR}\sqrt{P_s}s_x + \mathbf{H}_{SI}\sqrt{P_r}\mathbf{r}_x + \mathbf{w}_r, \tag{1}$$

where P_s and P_r are the transmit power constraint at the source and the relay, respectively, $\mathbf{w}_r$ is the additive white Gaussian noise (AWGN) with $\mathcal{CN}(0, \sigma^2)$.

Since the transmitted signal $\mathbf{r}_x$ is known at the relay, the received SI component $\mathbf{H}_{SI}\mathbf{r}_x$ can be subtracted after proper estimation from the received signal $\mathbf{r}_y$. However, due to the inherent non-idealities in the transmit RF-chain, a complete cancellation of SI is not possible. Canceling the SI component after subtracting the estimated SI from r_y, the received signal at the relay in (1) can now be written as:

$$\mathbf{r}_y = \mathbf{h}_{SR}\sqrt{P_s}s_x + \mathbf{n}_r + \mathbf{w}_r, \tag{2}$$

$\mathbf{n}_r$ denotes the RSI, whose elements are $n_r \sim \mathcal{CN}(0, \sigma^2_{rsi})$ where, $\sigma^2_{rsi} = \kappa P_r^{\upsilon}$ and κ and υ ($0 \leq \upsilon \leq 1$) are constants depending upon SI cancellation technique [10].

Upon receiving the signal from the source, the relay use AF scheme to forward the following signal,

$$\mathbf{r}_x = \beta r_y, \tag{3}$$

where β is the power amplification factor to ensure that the average power of the signal $\mathbf{r}_x$ satisfy the power constraint,

$$E\left[|\mathbf{r}_x|^2\right] = \beta^2\left(|h_{SR}|^2 P_s + \sigma^2_{RSI} + \sigma^2\right) \tag{4}$$

The received signal d_y at the destination can be written as

$$d_y = \mathbf{h}_{RD}\sqrt{P_r}\mathbf{r}_x + w_d, \tag{5}$$

where w_d is the AWGN noise with $\mathcal{CN}(0, \sigma^2)$.

Therefore, the end-to-end SINR from the source-to-destination via relay link can be expressed as [11]:

$$\Gamma_{SINR} = \frac{\Gamma_{SR}\Gamma_{RD}}{\Gamma_{SR} + (\Gamma_{RD} + 1)(\Gamma_{SI} + 1)}, \tag{6}$$

where

$$\begin{aligned} \Gamma_{SR} &= P_s |h_{SR}|^2 / \sigma^2, \\ \Gamma_{RD} &= P_r |h_{RD}|^2 / \sigma^2, \\ \Gamma_{SI} &= \sigma^2_{rsi}/\sigma^2. \end{aligned} \tag{7}$$

The SNR of channel link h_{SR} and h_{RD} are exponentially distributed with mean λ_{SR} and λ_{SR} respectively. So, the average channel SNR can be expressed as

$$\begin{aligned} \lambda_{SR} &= (1-\eta)N_R P_s d_{SR}^{-\nu}/\sigma^2, \\ \lambda_{RD} &= \eta N_R P_r d_{RD}^{-\nu}/\sigma^2, \end{aligned} \tag{8}$$

where ν denotes the pathloss exponent, $d_{SR} = D_{SR}/D = (1-\rho)D$ and $d_{RD} = D_{RD}/D = \rho D$ are normalized distances, ρ is a constant signifying the relative distance of relay with $0 < \rho < 1$.

III. SE and EE Evaluation

In this section, we formulate the SE and EE for FD relay system and then analyze the optimal antenna allocation scheme. The EE can be defined in terms of bits/Joule/Hz as the ratio of SE over the total transmitted power consumed, P_{tot} [23].

$$EE = \frac{SE}{P_{tot}} = \frac{SE}{P_s + P_r + P_{cir}}, \tag{9}$$

where P_s, P_r is the power transmitted by the source and the relay respectively and P_{cir} is the circuit power consumption.

The major power consuming device in the RF-circuitry is the high power amplifier (HPA), whereas the other RF components such as low-noise amplifier (LNA), mixers, filter, etc. consume far less power as compared to HPA hence can be ignored while calculating P_{tot} [24]. Further without any loss of generality, we can safely assume that the power consumed in each of the active RF transmit-chain (P_c) is identical [25]. Hence, $P_{cir} = (\eta N_r + 1)P_c$, where ηN_R is the total number of active transmit RF-chains in the FD relay.

Finally, the EE of the FD relay system can be formulated as:

$$EE = \frac{SE}{P_s + P_r + (\eta N_R + 1)P_c}, \tag{10}$$

where the SE is defined as:

$$SE = \log_2(1 + \overline{\Gamma}_{SINR}). \tag{11}$$

$$\overline{\Gamma}_{SINR}(\eta) = \frac{(1-\eta)N_R P_s (1-\rho)^{-\nu} D^{-\nu} \eta N_R P_r \rho^{-\nu} D^{-\nu}}{(1-\eta)N_R P_s (1-\rho)^{-\nu} D^{-\nu}\sigma^2 + (\eta N_R P_r \rho^{-\nu} D^{-\nu} + \sigma^2)(\sigma^2_{rsi} + \sigma^2)} \tag{12}$$

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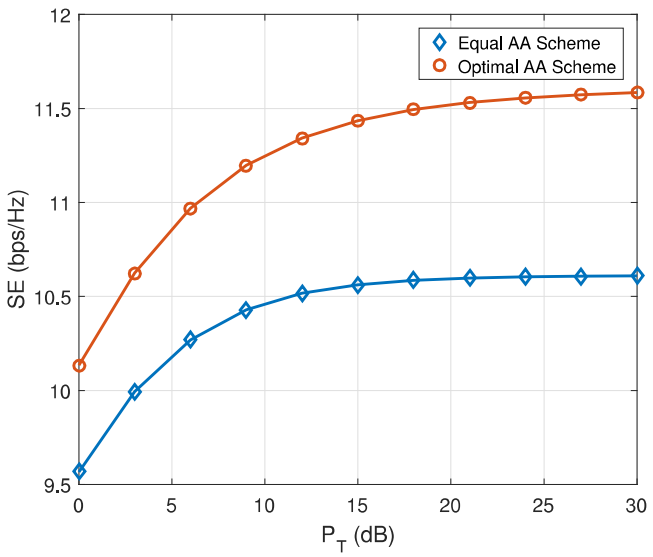

Fig. 3. Analytical (solid lines) and simulation (marker points) shows the SE of FD relay system with respect to transmit power P_T for both equal and optimal AA schemes at $\rho = 0.2$.

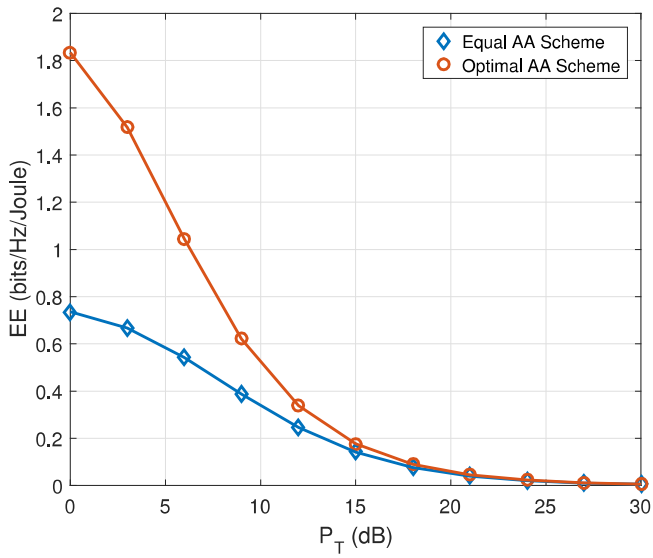

Fig. 4. Analytical (solid lines) and simulation (marker points) shows the EE of FD relay system with respect to transmit power P_t for both equal and optimal AA schemes at $\rho = 0.2$.

In order to maximize the SE, the SINR should be maximized as can be viewed from (11). Now from (6), (7) and (8), the average SINR ($\overline{\Gamma}_{SINR}$) is a function of η which can be expressed as shown in (12) (on the bottom of the last page).

Denoting $N_R P_s (1-\rho)^{-\nu} D^{-\nu}$ as a_1, $N_R P_r \rho^{-\nu} D^{-\nu}$ as a_2 and σ_{rsi}^2 as b, (12) can be re-written as:

$$\overline{\Gamma}_{SINR}(\eta) = \frac{(1-\eta)a_1 \eta a_2}{(1-\eta)a_1\sigma^2 + (\eta a_2 + \sigma^2)(b+\sigma^2)}, \quad (13)$$

The optimal η, that maximizes the average SINR can be evaluated as:

$$\partial \overline{\Gamma}_{SINR}(\eta)/\partial \eta = 0 \quad (14)$$

Solving (14) results in a second-order quadratic equation in η:

$$c_1 \eta^2 - 2c_2 \eta + c_2 = 0, \quad (15)$$

where $c_1 = a_1 - 2a_2 b - a_2 + b$ and $c_2 = a_1 + b + 1$. The roots of (15), i.e., $\eta = \left(c_2 \pm \sqrt{c_2(c_2 - c_1)}\right)/c_1$, maximizes the average SINR consequently the SE and the EE both are enhanced.

IV. Simulation Results

In this section, we discuss and present the simulation results for the performance of the proposed AA scheme. The simulation parameters are chosen as: $N_r = 20$, $\sigma^2 = -20$ dB, $\kappa = 0.01$, $\upsilon = 1$, $P_s = P_r = P_T$, $\nu = 2$, and $P_c = 0$ dB.

Fig. (3) shows the SE of the FD relay system with respect to transmit power at $\rho = 0.2$. The solid lines show the analytical curve, whereas the marker denotes the simulation results. The simulation results are in good agreement with the analytical results, which corroborates the proposed analysis. The result shows that the SE of the FD relay system can be enhanced through the proposed optimal AA scheme. As seen from the figure, the SE at $P_T = 15$ dB is 11.45 bps/Hz for the optimal AA scheme, whereas for equal AA scheme, it is only 10.05 bps/Hz. The result shows that the SE of the FD relay system saturates at higher transmit power, which is caused due to the impact of RSI at the FD relay node. Fig. (4) shows the EE of the FD relay system with respect to transmit power at $\rho = 0.2$. As evident from the result, the EE of the FD system can be enhanced significantly through a optimal AA scheme. For instance, the EE for optimal AA scheme at $P_T = 5$ dB is 1.2 bits/Hz/Joule, whereas for the same transmit power, the EE of equal AA scheme is only 0.6 bits/Hz/Joule.

Fig. (5) shows the SE and EE with respect to the localization parameter ρ for a zero RSI FD relay system, i.e., complete SI cancellation is happening at the FD relay. The result shows that for $\rho = 0.5$, maximum SE and EE are achieved at $\eta = 0.5$, whereas for $\rho \neq 0.5$, $\eta = 0.5$ is not an optimal AA parameter. As evident from the result, the SE and EE both can be enhanced through optimal AA scheme. It can also be noted here that as shown in the figure, for a zero RSI based FD relay system, the optimal location of the relay is at $\rho = 0.5$, i.e., at the mid-point of the source-to-destination link which is quite obvious.

Fig. (6) shows the SE and EE with respect to the localization parameter ρ of the FD relay system impaired by RSI. The results show that the proposed optimal AA scheme outperforms the traditional equal AA scheme, as evident from Fig. (6). For instance at $\rho = 0.4$, the SE is 11 bps/Hz for the equal AA scheme, while the proposed AA scheme enhances it to 12 bps/Hz. Similarly at $\rho = 0.4$, the EE for the equal AA scheme is 0.027 bits/Hz/Joule, whereas the proposed AA scheme enhances it to 0.030 bits/Hz/Joule. The SE and EE of a FD system show interesting trend with respect to the AA scheme when the position of the relay is varied. As shown in

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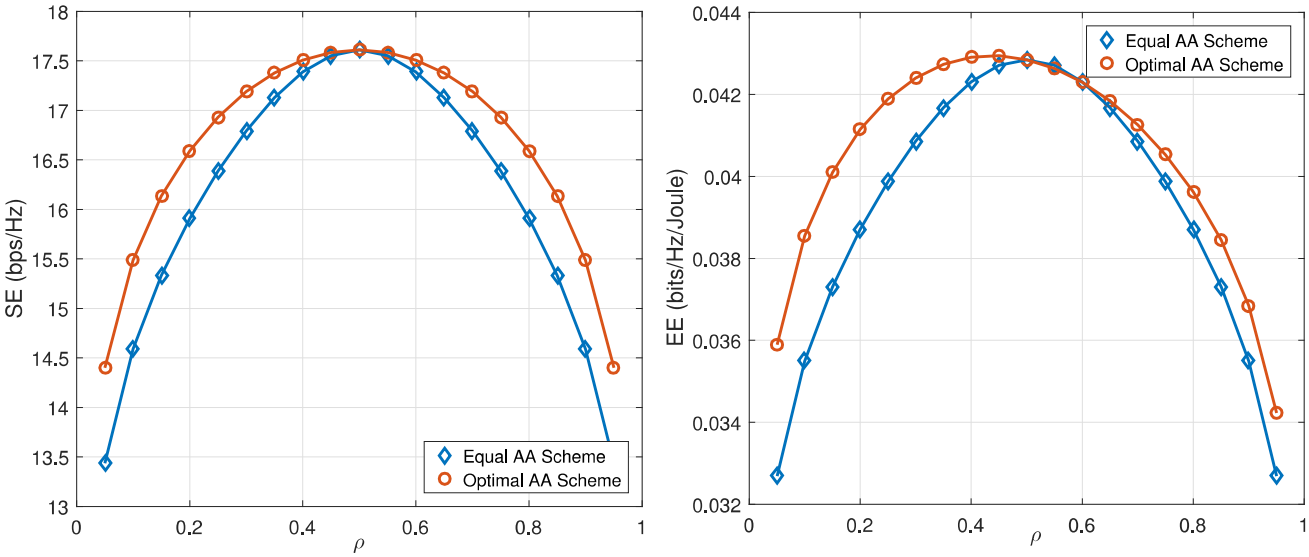
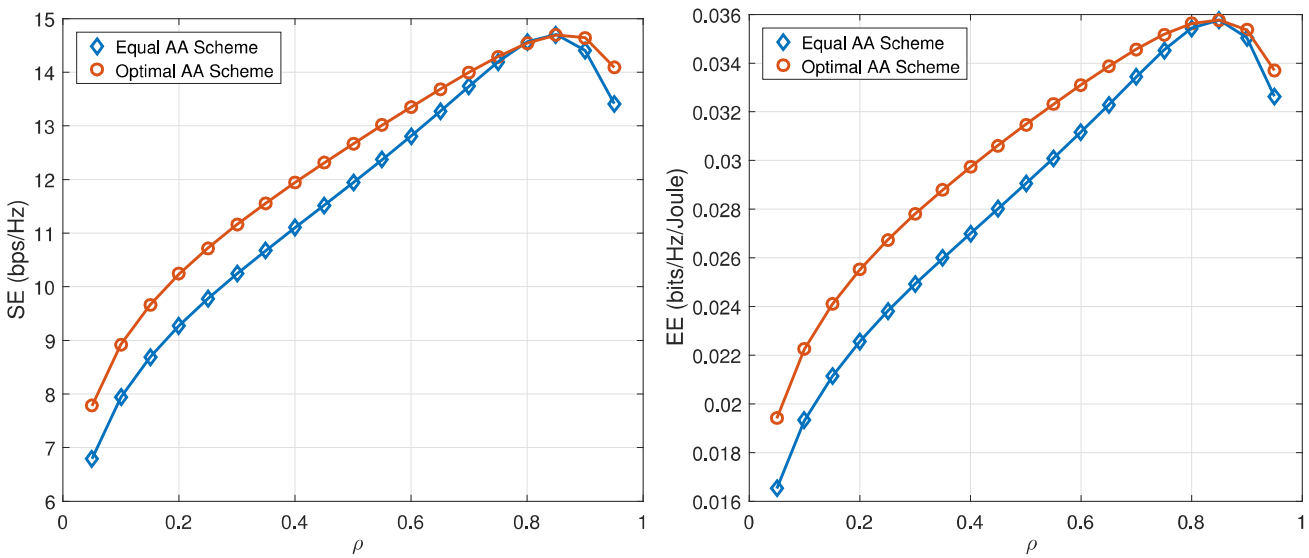


(a) SE (b) EE

Fig. 5. Analytical (solid lines) and simulation (marker points) shows the SE and EE with respect to distance parameter ρ for both equal AA and optimal AA schemes of a FD relay system at transmit power $P_T = 20$ dB without RSI.



(a) SE (b) EE

Fig. 6. Analytical (solid lines) and simulation (marker points) shows the SE and EE with respect to distance parameter ρ for both equal AA and optimal AA schemes of a FD relay system at transmit power $P_T = 20$ dB with RSI.

Fig. (6), initially, both SE and EE increases with ρ for both the AA schemes, viz optimal and equal AA scheme, till $\rho = 0.85$. However, for $\rho > 0.85$ the SE and EE performance of the AA schemes deteriorates slightly. This is due to the fact that after $\rho = 0.85$, the relay node is too far from the destination and hence has larger path loss.

The results also show that the localization factor ρ plays an important role in resource allocation in a practical FD relay system impaired by RSI. It can be noted here that the SE and EE are maximum for $\rho = 0.5$ for an ideal FD relay system with complete SI cancellation, i.e., zero RSI. In contrary to that, for a practical FD system impaired by RSI, $\rho = 0.5$ is not an optimal location, as evident from Fig. (5) and Fig. (6). For a practical FD relay system, $\rho = 0.85$ maximizes the SE and EE, which implies that the relay should be much closer to the source as compared to the destination. This can be attributed to the fact that the received signal strength at the relay increases with ρ due to reduced pathloss, as compared to the interference due to RSI, which remains constant as it depends on relay transmit power which is kept at $P_T = 20$

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dB. The result also verifies the proposed optimal AA scheme outperforms the equal AA schemes for both ideal as well as practical FD relay system. The close agreement between the analytical and simulation results further corroborate the proposed analysis.

V. Conclusion

In this work, we analyzed the SE and EE of a FD relay system with respect to the proposed AA scheme at the FD relay node. In the proposed AA scheme, the antennas are optimally distributed for transmission and reception at the FD relay, as the source-to-relay link is impaired by the RSI, whereas relay-to-destination link is free from interference. It has been shown that the proposed optimal AA scheme enhances the SE and the EE of the FD relay system as compared to the equal AA scheme. Further, the impact of relay location has also been investigated. It has also been shown that the relay location impacts the SE and EE of the FD relay system. The SE and EE can be maximized through optimal AA scheme and relay location parameter. The simulation results are in good conformity with the analytical results, validating the proposed analysis.

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